

Minority athletic performance, racial attitudes, and racial hate

Preliminary results, please do not cite

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Football in America

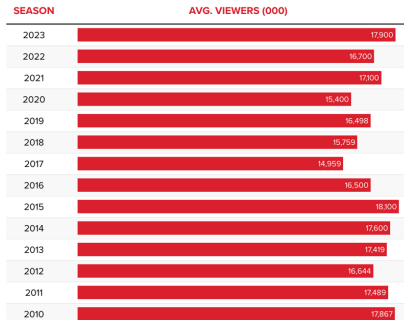
It's a big deal



Football in America

- ▶ The Super Bowl is consistently the most viewed event on US television (ratings typically over 40%)
- ▶ Even regular season NFL games are highly watched, averaging over 15 million viewers
- ▶ Important to national and local identity

NFL Regular-Season Viewership Trend



Quarterbacks in the NFL

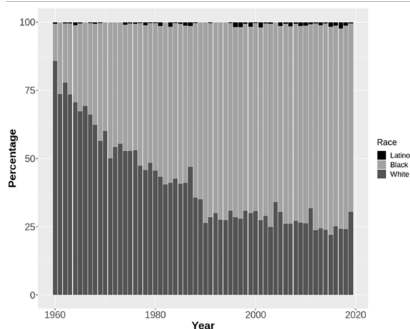
Often viewed as the “Face of the Franchise”:

- ▶ widely viewed as the most important position in football
- ▶ lead the offense by calling plays, executing strategy, and making split-second offensive play decisions
- ▶ symbols of leadership, excellence, and success
- ▶ generally the highest earning position

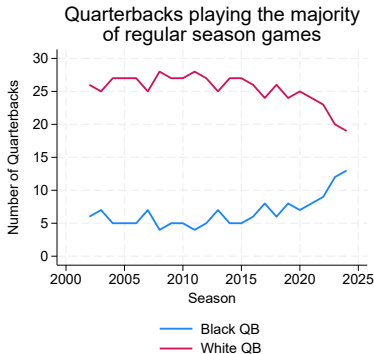
Black quarterbacks in the NFL

Today many quarterbacks are Black but until recently they were rare

- ▶ Since 1990, the Black player share in the NFL has been consistently over 60%, but between 2000 and 2020 only 20% of quarterbacks were Black



1. Percentage of professional football players by race/ethnicity from 1960 to 2020.



Black quarterbacks' impacts on fans

“If the team won and Moon played well, he was the Oilers’ quarterback. If Moon played poorly and Houston lost, he was its Black quarterback. Or worse. Much worse.”

– Jason Reid on Hall of Famer Warren Moon

How do star minority athletes, like Black quarterbacks, impact racial attitudes and racial hate?

Do these impacts depend on their success or failure?

This paper:

- ▶ **Measuring Racial Bias:**

- ▶ Hard outcomes: hate crimes
- ▶ Expressive indicators: hate speech on social media, racial-slur Google searches, implicit-bias indices

- ▶ **Analytical Framework:**

- ▶ Medium run – evolution of attitudes over the course of the season and beyond
- ▶ Short run – emotional, immediate responses

- ▶ **Findings:**

- ▶ Medium run: Hate crimes and Google searches for racial slurs decline when Black quarterbacks' teams reach the championship game.
- ▶ Short run: presence of Black quarterbacks leads to larger spikes in hate crimes and hate-speech activity after losses, and relatively less hate speech after wins

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Literature and contribution

Emotions, sports and violent behavior (Rees and Schnepel 2009, Card and Dahl 2011)

Differences in rewarding/punishing of in/out-groups (Sarsons (2017), Egan et al. (2022)), Skiba et al. 2002, Gregory et al. 2010)

Minority representation in leadership and role model positions.

- ▶ Generally focused on academic and aspirational outcomes of minority groups (Gershenson et al. (2022), Lim and Meer (2020), Fairlie et al. (2014), Dee (2005),...)
- ▶ Little on media exposure to celebrity role-models on majority group prejudice (Paluck et al. (2021), Kearney and Levine (2020), Bertrand and Duflo (2017))
- ▶ Famous minority role models (President Obama, the soccer star Mohamed Salah) reduced prejudice (Plant et al. 2009, Marble et al. 2021)

We move beyond the study of individual minority role-models, to consider whether these patterns are systematic, generalizable

We separately explore reactions to players' successes and failures.

We use expressive and hard measures of racial bias, and examine their dynamic patterns across different time horizons.

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Outcome data

FBI hate crimes (using 2002-2024):

- ▶ Hate crime incidents collected by FBI's Uniform Crime Reporting (UCR) data (widely believed to be an underestimate of actual incidents)

Implicit association test scores (2002-2024):

- ▶ psychological tests designed to measure implicit biases thru rapid sorting exercises
- ▶ available online thru *Project Implicit*

Reddit submissions and comments (2009-2024):

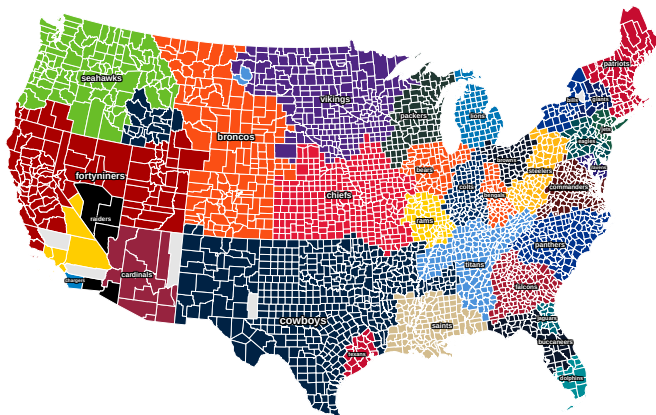
- ▶ collect all submissions and comments posted to NFL team subreddits and city subreddits for 100 largest cities.
- ▶ for each measure hate speech and sentiment using standard packages (HATEBERTA for hate speech, VADER for sentiment)

Racial Animus Index (googles of racial slur) (season only 2004-2023):

- ▶ build from Google Trends indexes for searches of racial slurs in DMA's over 2 month periods
- ▶ Stephens-Davidowitz (2014), Chetty et al. (2022), Kline et al. (2022), Derenoncourt (2022)

Team affiliations

Locations are mapped to NFL teams using ratings and google search index data [Method](#)



Quarterback Data

Collected NFL play-by-play game data and player data using public APIs for 2002-2024 seasons and had RA's classify race of all quarterbacks [Details](#)

Game level variables (10,817 $game \times teams$):

- ▶ use games that are min 7 days after previous game and min 4 days before next
- ▶ $Win_{gn} = 1$ if team n won game g . 0 if they lose or tie (rare)
- ▶ $QBcomp_{gn} \in \{WW_{=6560}, WB_{=1854}, BW_{=1859}, BB_{=544}\}$: The QB racial composition (local, opp) using the QB for each team that had the most game time

Season level variables (736 $season \times teams$):

- ▶ $QBBlack_{ns} = 1$ if the majority of regular season games had a Black QB playing the majority of the game (152 $season \times teams$)
- ▶ $Perf_{ns} \in \{NP_{=450}, NC_{=194}, C_{=92}\}$ team performance is classified as not making the playoffs (NP), making the playoffs but not championships (NC), or making the championships (C)

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Season specification

$$Y_{Inst} = \beta_0 + \sum_{m \neq 1}^5 \beta_{1m} 1\{MoBin_t = m\} \times QBBlack_{ns} + \lambda_{ls} + \mu_{ml} + \psi_{Pm} + \epsilon_i$$

$\hat{\beta}_{1m}$'s estimate the difference in outcomes for Black (relative to white) QB teams for two month periods with fixed effects for

- ▶ *location-season* (λ_{ls}): restricts to within location-season variation relative to omitted period (Jul-Aug)
- ▶ *month-location* (μ_{ml}): controls for l specific seasonality
- ▶ *performance-month* (ψ_{Pm}): controls for performance induced seasonality for the omitted group (white QB teams)

Also estimate $\hat{\beta}_{1m}^P$: difference for different performance levels.

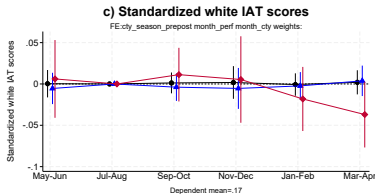
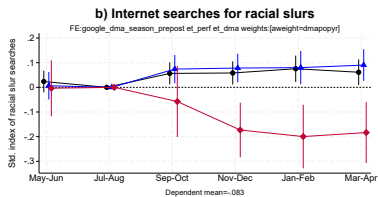
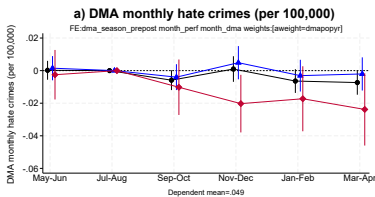
Team quarterback selection on race? Little evidence of selection that would affect within-season estimates [Table](#)

Season: Anti-Black attitudes and hate

In areas with championship level Black QBs teams: In levels

- ▶ Greater ↓ in anti-black hate crimes (vs. smaller ↓)
- ▶ ↓ searches for racial slurs (vs. none for white QB teams)
- ▶ No increases in IAT scores (vs. ↑ for white QB teams)

No change in slur searches in non-playoff areas (vs. ↓ for white QB areas)

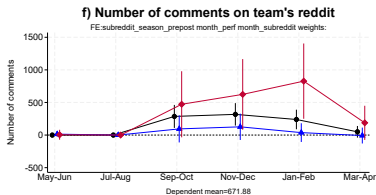
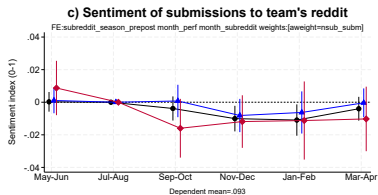
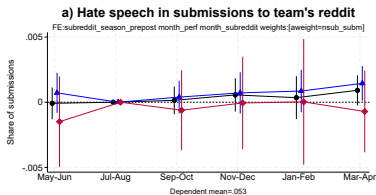


- Black v. White QB difference - All teams
- Black v. White QB difference - Teams not in playoffs
- Black v. White QB difference - Teams in playoffs but not championships
- Black v. White QB difference - Teams in championship

Season: Behavior on team reddit

On higher performing team reddit with Black QBs: In levels

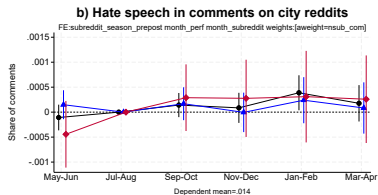
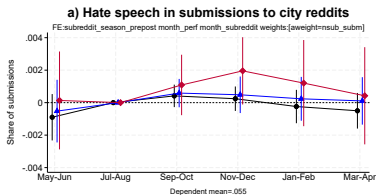
- ▶ Little impact on hate speech
- ▶ More negative sentiment relative to white QB teams at same performance level
- ▶ More activity, especially comments



- Black v. White QB difference - All teams
- ▲— Black v. White QB difference - Teams not in playoffs
- Black v. White QB difference - Teams in playoffs but not championships
- ◆— Black v. White QB difference - Teams in championship

Season: Hate speech on city reddit

No strong evidence of differences in hate speech on city reddit In levels



- Black v. White QB difference - All teams
- Black v. White QB difference - Teams not in playoffs
- Black v. White QB difference - Teams in playoffs but not championships
- Black v. White QB difference - Teams in championship

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Game specification

$$\begin{aligned} Y_{ignd} = & \beta_0^{WW} + \sum_{d \neq -1}^7 \beta_{1d}^{WW} 1\{Day_{dg} = d\} + \sum_{C \neq WW}^3 \sum_{d \neq -1}^7 \beta_{2d}^C 1\{Day_{dg} = d\} \times 1\{QBcomp_{gn} = C\} \\ & + \sum_{d \neq -1}^7 \beta_{3d}^{WW} 1\{Day_{dg} = d\} \times Win_{gn} + \sum_{C \neq WW}^3 \sum_{d \neq -1}^7 \beta_{4d}^C 1\{Day_{dg} = d\} \times 1\{QBcomp_{gn} = C\} \times Win_{gn} \\ & + \gamma_{lgn} + \omega_w + \epsilon_i \end{aligned}$$

- ▶ $\hat{\beta}_{2d}^C$: estimates the difference in outcomes for days around **lost** games with composition C (relative to WW QB lost games)
- ▶ $\hat{\beta}_{2d}^C + \hat{\beta}_{4d}^C$: estimates the difference in outcomes for days around **won** games with composition C (relative to WW QB won games)

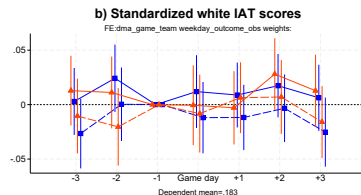
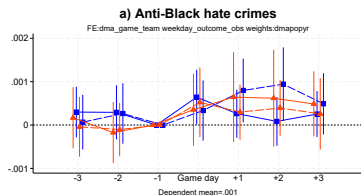
Controlling for:

- ▶ *location-game-team* (γ_{lgn}): restricts to within location-game-team variation relative to the day before the gameday
- ▶ *weekday* (ω_w): day of the week controls

Game: Anti-Black attitudes and hate

As compared to games where both QBs are white: In levels

- ▶ Some evidence of $\uparrow$ in anti-black hate crimes when lose with local BQB
- ▶ Some evidence of $\uparrow$ anti-black hate crime when opp. QB is Black
- ▶ More pronounced effects for unexpected game outcomes By expectation
- ▶ No evidence of effects on IATs



Wins

Losses

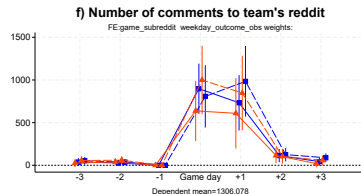
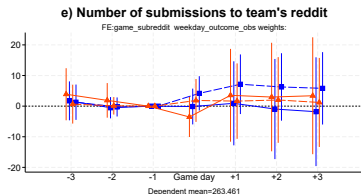
Diff. WW and BW

Diff. WW and WB

Games: Traffic on team reddit

As compared to games where both QBs are white: In levels

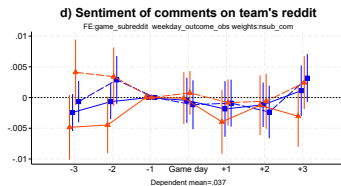
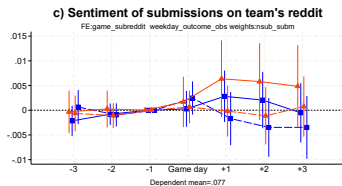
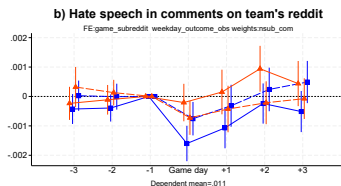
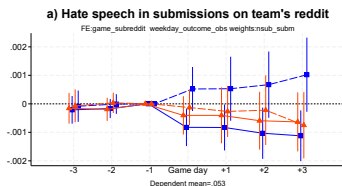
- ▶ no major differences in submission volume
- ▶ substantially more comments (25% ↑ when any BQB is on the field)



Games: Behavior on team reddit

As compared to games where both QBs are white: In levels Expectations

- ▶ less ↑ in hate speech when win with local Black QB
- ▶ more ↑ in hate speech (submissions only) when loose with local BQB
- ▶ more ↑ in submission sentiment when win against a opp. BQB



Wins —■—

Losses - -■-

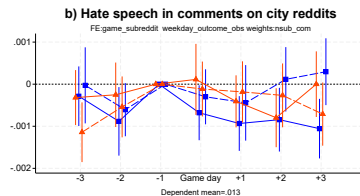
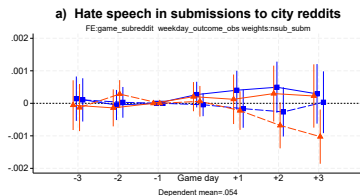
Diff. WW and BW

Diff. WW and WB

Games: Hate speech on city reddit

As compared to games where both QBs are white: In levels

- No evidence of differences in hate speech on city reddit



Wins		
Losses		
Diff. WW and BW		
Diff. WW and WB		

Thank you!
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Mapping media markets to NFL teams [Back](#)

For each of the 32 teams and each media market (DMA) we calculate

$$Rating_{DMA}^{team} = \overline{HomeRatings}^{team} \times SearchIndex_{DMA}^{team},$$

- ▶ $\overline{HomeRatings}^{team}$ is average reported ratings in a teams' local DMA for the five seasons this data is available (2011, 2012, 2013, 2020, 2021)
- ▶ $SearchIndex_{DMA}^{team}$ is the 2004-2025 Google trends interest index measure of DMA searches for each team.
- ▶ DMA's are matched to their highest rated team

Team quarterback selection on race Back

Do teams consider fans changing racial attitudes in their selection of QBs?

Little evidence that within team changes in fan hate speech predict QB race

Little evidence that within team changes in fan hate speech predict quarterback race

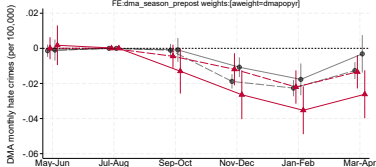
	(1)	(2)	(3)	(4)	(5)
	Main QB is Black				
Panel a: Level of Reddit hate speech predicting quarterback race					
Std. hate speech last season	-0.228*** (0.0868)	-0.0598 (0.0915)	-0.0737 (0.102)	-0.0688 (0.0923)	-0.0627 (0.0919)
Last season QB was Black				0.428*** (0.0499)	0.264*** (0.0594)
Last season made playoffs only with Black QB					0.358*** (0.0913)
Last season made championships with Black QB					0.508*** (0.134)
Last season made playoffs only					-0.0635 (0.0452)
Last season made championships					-0.0790 (0.0647)
Panel b: Changing levels of Reddit hate speech predicting quarterback race					
Std. change in hate speech from s-2 to s-1	0.000915 (0.0214)	-0.00345 (0.0186)	-0.00958 (0.0196)	-0.00444 (0.0179)	-0.0106 (0.0176)
Last season QB was Black				0.428*** (0.0500)	0.261*** (0.0596)
Last season made playoffs only with Black QB					0.361*** (0.0915)
Last season made championships with Black QB					0.516*** (0.134)
Last season made playoffs only					-0.0664 (0.0450)
Last season made championships					-0.0823 (0.0641)
Observations	392	389	389	389	389
FE: Subreddit	No	Yes	Yes	Yes	Yes
FE: Season	No	No	Yes	Yes	Yes

Season in levels: Anti-Black attitudes and hate

Differences

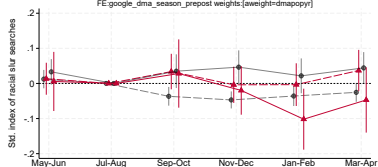
a) DMA monthly hate crimes (per 100,000)

FE:dma_season_prepost weights:[aweight=dmapopyr]



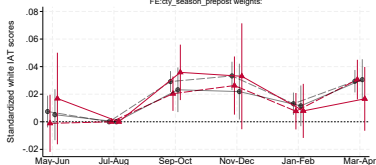
b) Internet searches for racial slurs

FE:google_dma_season_prepost weights:[aweight=dmapopyr]



c) Standardized white IAT scores

FE:cty_season_prepost weights:



Main QB is white

Main QB is Black

All

Not in playoffs

In championships

Season: Behavior on team reddit's Differences

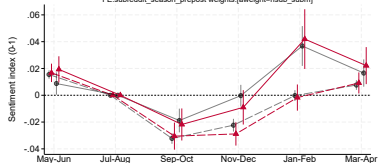
a) Hate speech in submissions to team's reddit

FE:subreddit_season_prepost weights:[aweight=nsbm]



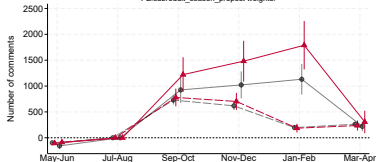
c) Sentiment of submissions to team's reddit

FE:subreddit_season_prepost weights:[aweight=nsbm]



f) Number of comments on team's reddit

FE:subreddit_season_prepost weights:

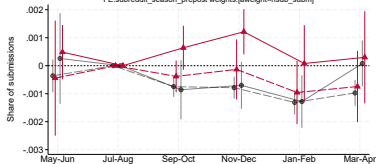


Season: Hate speech on city reddit

Differences

a) Hate speech in submissions to city reddit

FE:subreddit_season_prepost weights:[weight=nsub_subm]



b) Hate speech in comments on city reddit

FE:subreddit_season_prepost weights:[weight=nsub_com]



Main QB is white

Main QB is Black

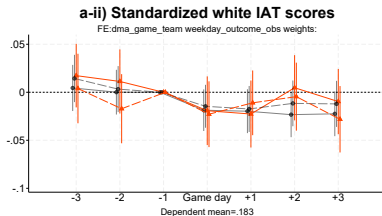
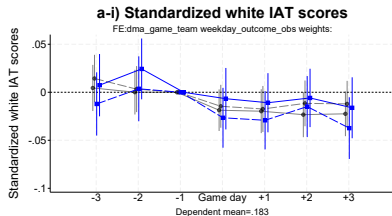
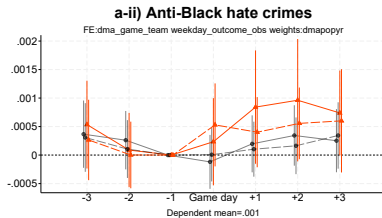
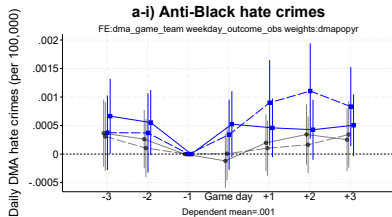
All

Not in playoffs

In championships

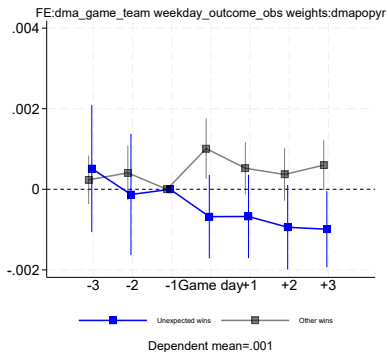
Game: Anti-Black attitudes and hate

Differences

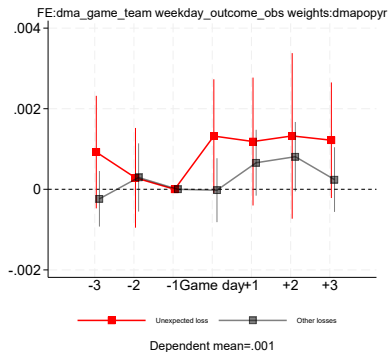


a) Anti-Black hate crimes

i - Wins



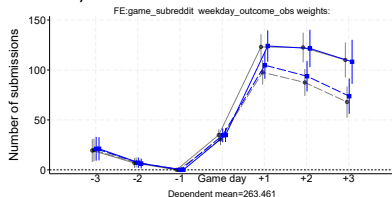
ii - Losses



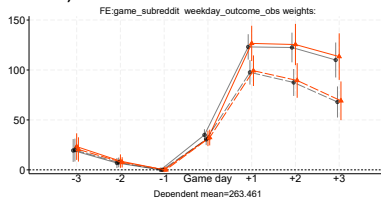
Games: Traffic on team reddit

Differences

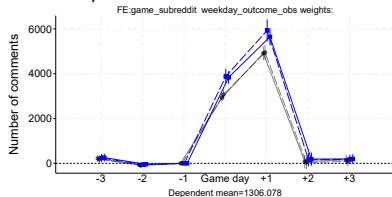
a-i) Number of submissions to team's reddit



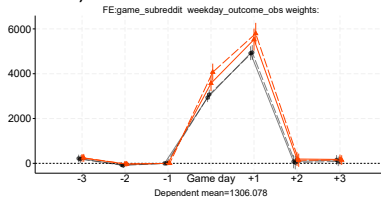
a-ii) Number of submissions to team's reddit



b-i) Number of comments to team's reddit



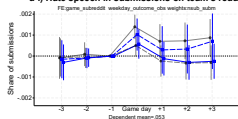
b-ii) Number of comments to team's reddit



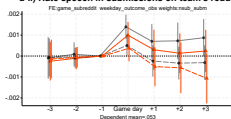
Games: Behavior on team reddit's

Differences

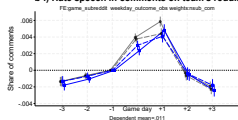
a-i) Hate speech in submissions on team's reddit



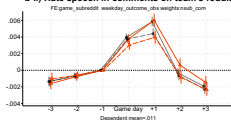
a-ii) Hate speech in submissions on team's reddit



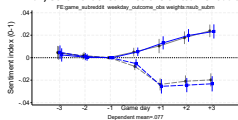
b-i) Hate speech in comments on team's reddit



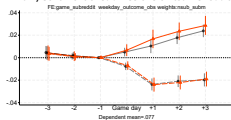
b-ii) Hate speech in comments on team's reddit



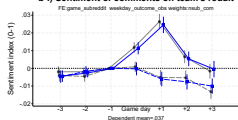
a-i) Sentiment of submissions on team's reddit



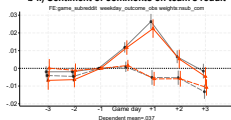
a-ii) Sentiment of submissions on team's reddit



b-i) Sentiment of comments on team's reddit



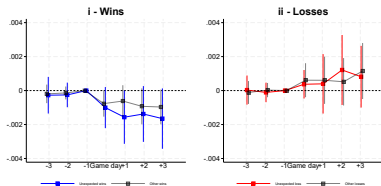
b-ii) Sentiment of comments on team's reddit



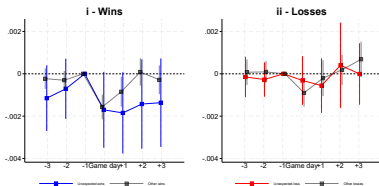
Games and expectations: Hate speech

[Back](#)

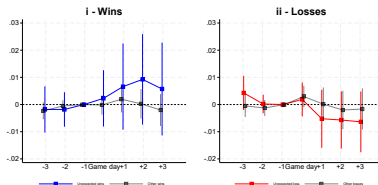
a) Hate speech in submissions on team's reddit



b) Hate speech in comments on team's reddit



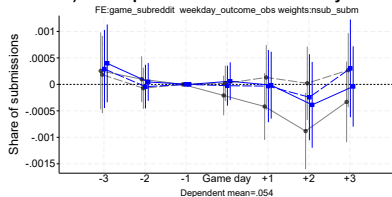
c) Sentiment of submissions on team's reddit



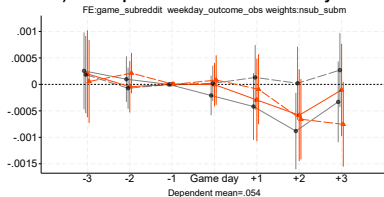
Games: Hate speech on city reddit

Differences

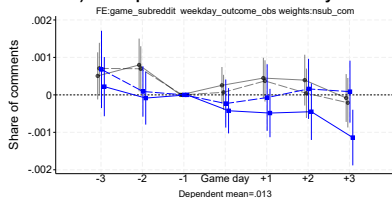
c-i) Hate speech in submissions to city reddit



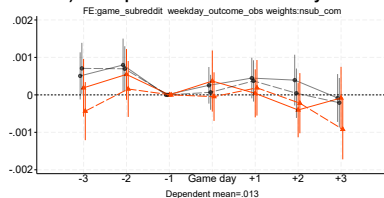
c-ii) Hate speech in submissions to city reddit



d-i) Hate speech in comments on city reddit



d-ii) Hate speech in comments on city reddit



NFL game and player data [Back](#)

We collected NFL game and player data using public APIs from ESPN.com and NFL.com.

- ▶ The datasets cover 4,806 games during the 2002–2019 seasons (4608 Regular season, 198 post season).
- ▶ The game data includes detailed information on each play, along with the players involved.
- ▶ The player data contains each player's name, team, position, and other season-level attributes.
- ▶ We merged the game data with the player data to link plays with player characteristics.
- ▶ With the assistance of several research assistants, we manually identified the race of each quarterback in the dataset.
- ▶ In total, 49 Black QBs among 243 QBs.
1677 games out of 4806 were led by Black QBs.